PANCHATANTRA AI

Dataset Identification, Age Suitability Framework, MVP and System Design Documentation

Research and Prototype / MVP
Indian Knowledge Systems, AI, RAG, Adaptive Learning and Generative Multimedia

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Prepared for academic research and prototype development

1. Executive Summary

Panchatantra AI is an intelligent adaptive storytelling platform designed to make Panchatantra stories more engaging for children while preserving the cultural and moral context of the original narratives. The platform combines a curated Panchatantra story corpus, Retrieval-Augmented Generation (RAG), large language models, age-aware adaptation, quizzes, moral reasoning and multimedia storytelling.

A key MVP addition is Parent Voice Story Mode. A parent creates and controls a private voice profile, then selected Panchatantra stories can be narrated using that parent's voice. The child receives an AI-generated narration rather than a generic narrator. The voice feature is consent-based, private and restricted to the parent owner.

IGNCA documents a Panchatantra exhibition involving 160 children, grouped as below 7, 8-12 and 13-17 years. It reports children's interpretations of stories including Monkey and the Crocodile, Lion and the Hare, and Hare and the Partridge. This is useful evidence for studying story interpretation by age rather than assigning age labels only by intuition. □cite□turn0search0□turn0search24□

2. Problem Statement

Children often encounter Panchatantra through static books, short retellings, videos or generic narration. These formats do not necessarily adapt to reading level, prior learning or preferred listening experience. An unrestricted LLM can also change events or morals. The research problem is: How can AI adapt Panchatantra stories to different children's reading and reasoning levels while preserving the authenticity of the underlying Indian Knowledge System?

3. Objectives

- Build a traceable Panchatantra story corpus with source references.
- Develop a multi-dimensional age-suitability framework.
- Use RAG so adaptations remain grounded in source material.
- Generate age-appropriate stories, quizzes and moral-reasoning questions.
- Implement Parent Voice Story Mode as an MVP.
- Evaluate factuality, age suitability, cultural fidelity, comprehension and voice quality.

4. Dataset Identification and Data Strategy

The dataset should be layered rather than one large scraped collection. Source text, annotations and generated adaptations must remain separate so every output can be traced to its source.

Layer	Purpose	Key fields
A. Story Corpus	Source-of-truth narratives	story_id, title, tantra, source, original_text, translation, provenance
B. Story Annotation	Content and learning properties	characters, theme, moral, length, complexity, sensitivity
C. Age Suitability	Research age labels	language, plot, moral, length, sensitivity, validator_rating
D. Learning Dataset	Quiz and reasoning evaluation	question, type, answer, skill, difficulty
E. Adaptation Dataset	Generated versions	source_story_id, target_age, prompt_version, output, review_status
F. Voice Metadata	Private parent voice control	voice_id, owner_id, consent, language, timestamps

4.1 Source Strategy

IGNCA is a strong institutional research source because it provides Panchatantra context and an analysis of children's responses. The source identifies several stories and records how children connected them with different morals. □cite□turn0search0□ Modern children's websites may be used for comparison, but should not automatically be treated as the source of truth. Prefer public-domain, institutionally hosted or appropriately licensed source texts and record provenance for every story.

4.2 Initial Story Set

Story	Preliminary band	Research reason
Lion and the Hare	6-9	Short plot, clear conflict, intelligence and courage.
Crow and Snake	6-9	Cause-effect reasoning and problem solving.
Monkey and the Crocodile	8-10	Dialogue, friendship, jealousy and presence of mind.
Four Friends	7-10	Cooperation and multiple characters.
Tortoise and Geese	7-10	Self-control and consequences.
Blue Jackal	8-11	Deception, identity and consequences.
Brahmin and Mongoose	9-12	Emotionally sensitive decision consequences.
Loss of Friends	11-15	Layered relationships and moral reasoning.

These age bands are preliminary research hypotheses, not final validated labels. They should be validated with independent annotators and teachers or child-literature experts.

5. Age Suitability Framework

The system must separate reading difficulty from moral difficulty. A story can have simple vocabulary but require advanced reasoning. Age suitability should therefore be multi-dimensional.

Dimension	What is checked	Scale
Language complexity	Vocabulary, sentence length, syntax	1-5
Plot complexity	Events, characters, cause-effect	1-5
Moral complexity	Concrete lesson vs abstract/conflicting values	1-5
Length	Word count and listening time	1-5
Emotional sensitivity	Fear, death, violence, grief or distress	1-5
Cultural context	Need for cultural/historical explanation	1-5
Comprehension demand	Inference, prediction, perspective taking	1-5

5.1 Validation Method

- Two independent annotators score each story.
- Create an annotation guide from disagreements.
- Ask teachers or child-literature experts to rate suitability.
- Measure inter-rater agreement and revise the rubric.
- Only then convert scores into recommended age ranges.
- Child evaluation should require appropriate institutional and parental consent.

The IGNCA study provides useful methodological evidence because responses were grouped by age and analyzed for specific stories. It does not prove that any one story has one fixed age range. □cite□turn0search0□

6. System Architecture

Layer	Components
Client	React/web or mobile UI, child profile, parent dashboard, story player
Application API	FastAPI or Node.js, authentication, story requests, progress
Knowledge	PostgreSQL metadata plus FAISS/Chroma vector store
AI	Embeddings, RAG retriever, LLM, adaptation and quiz generation
Multimedia	TTS, parent voice synthesis, illustrations, audio player
Evaluation	Factuality, source comparison, age rubric, quiz analytics

7. Complete Process Flow

Parent / Child Profile → Age + Learning History → Story Recommendation → RAG Retrieval → Story Analysis → Age Adaptation → Cultural Context Check → LLM Generation → Parent Voice / Standard TTS → Story Player → Quiz → Moral Reasoning → Performance Update → Next Recommendation

7.1 RAG Flow

User Request → Query Embedding → Vector Search → Relevant Story Chunks → Context + Target Age → LLM → Grounded Output → Source/Provenance Record → Fidelity Check

7.2 Story Generation Flow

Original Story → Retrieve Source → Extract Characters/Events/Moral → Determine Adaptation Level → Adapt Language → Add Cultural Explanation → Generate Questions → Generate Narration Script → Audio → Final Story

8. MVP Feature: Parent Voice Story Mode

The parent voice feature lets a parent record or upload their own voice, create a private voice profile and select that voice for Panchatantra narration. The AI still controls story adaptation; the parent voice controls the narration identity.

8.1 MVP Flow

Parent Login → Consent and Voice Rights Confirmation → Record Sample → Provider Verification / Voice Creation → Private Voice ID → Select Child Profile → Select Story → Age Adaptation → Narration Script → Parent Voice Synthesis → Story Player

8.2 Recommended MVP Implementation

- Use a managed voice-cloning provider rather than training a voice model from scratch.
- The parent creates and controls the voice profile. Do not clone a child's voice.
- Keep the voice ID separate from child profile data.
- Require authentication and authorization for every generation request.
- Provide deletion and revocation controls.
- For the first prototype, support one parent voice, one language and a small story set.

Current ElevenLabs documentation describes Instant Voice Cloning using about 1-2 minutes of clean audio and requires confirmation that the user has the right and consent to clone the voice. Its Professional Voice Cloning process includes verification and is designed for the creator's own voice. Therefore the MVP should make the parent the verified owner of the voice profile. □cite□turn0search3□turn0search1□turn0search7□

8.3 Voice Safety and Privacy

- Explicit consent before voice enrollment.
- No minor voice enrollment in the MVP.
- Private-by-default voice profiles.
- No public sharing by default.
- Encrypt recordings and provider identifiers.
- Allow voice deletion and revocation.
- Log voice creation and generation events.
- Never use the parent voice for unrelated messages or impersonation.
- Clearly label AI-generated narration in the parent dashboard.

Voice cloning can generate speech the original speaker never recorded. Provider documentation therefore emphasizes consent, verification and restrictions on unauthorized impersonation. The design should treat the voice profile as a sensitive asset. □cite□turn0search4□turn0search9□

8.4 Voice Data Flow

Parent Audio → Consent Check → Voice Provider → voice_id → Secure Metadata Store → Adapted Story Text → TTS Request → Generated Audio → Temporary Storage → Story Player → Retention/Deletion Policy

9. Data and Database Schema

9.1 Story Dataset Fields

Field	Description
story_id	Unique story identifier
title	Canonical title
source_reference	Source and provenance
original_text	Source text where legally usable
translation_en / translation_hi	Project translations
tantra	Panchatantra section
characters / themes	Structured annotations
moral / learning_objective	Source-supported learning annotations
complexity_scores	Age rubric values
sensitivity_flags	Content warnings
review_status	Pending / reviewed / approved

9.2 Parent Voice Fields

Field	Purpose
voice_id	Provider/model reference
parent_user_id	Voice owner
consent_version	Accepted consent text version
verification_status	Pending / verified / rejected
language	Narration language
created_at / deleted_at	Lifecycle timestamps
provider	Voice service

10. Evaluation Plan

Area	Metric / method	Purpose
Retrieval	Recall@K / expert relevance	Correct source retrieval
Factuality	Expert review / source comparison	Detect invented events or morals
Cultural fidelity	Expert rating	Preserve context
Age suitability	Teacher rating + rubric agreement	Validate age band
Comprehension	Quiz score	Measure understanding
Moral reasoning	Scenario-response rubric	Measure reasoning
Engagement	Completion / optional rating	Compare formats
Voice quality	Parent/listener rating	Naturalness and intelligibility

11. MVP Scope

Must have: 10-20 curated stories, provenance, age rubric, RAG, age-adapted text, standard narration, parent voice enrollment, parent voice narration, quizzes, child profile and basic progress.

Should have: English and Hindi, cultural explanation cards, basic illustrations, recommendations based on age and quiz performance.

Later: full animation, multilingual parent voices, advanced recommendation models, teacher dashboard and longitudinal learning studies.

12. Risks and Mitigations

Risk	Mitigation
LLM changes the moral	RAG, source comparison and expert review
Wrong age label	Multi-dimensional rubric and teacher validation
Copyright uncertainty	Provenance plus public-domain/licensed sources
Voice misuse	Verified parent ownership, consent, private access and deletion
Child privacy	Minimize personal data
Sensitive content	Content flags and age filtering
Poor voice quality	Clean, language-matched recordings and QA

13. Development Plan

Weeks 1-2: story collection, provenance and dataset setup.
Weeks 2-4: age rubric, independent annotation and teacher review protocol.
Weeks 4-6: embeddings, vector database, RAG and grounded generation.
Weeks 6-8: age adaptation, quizzes and moral reasoning.
Weeks 7-9: Parent Voice MVP, consent flow and narration API.
Weeks 9-12: multimedia player, testing and evaluation.

14. Team Responsibilities

Member	Responsibility
1	IKS research, story collection, source verification and annotation
2	RAG, embeddings, vector DB, prompts and factuality checks
3	Multimedia, TTS, Parent Voice MVP and audio pipeline
4	Frontend/backend, database, recommendation and evaluation

15. Expected Research Outputs

- Provenance-aware Panchatantra dataset.
- Age-suitability annotation framework.
- Evaluated RAG adaptive storytelling prototype.
- Parent Voice Story Mode MVP.
- Quiz and moral-reasoning evaluation set.
- Results on factuality, age suitability, comprehension and voice quality.
- Final research report or paper.

16. References

IGNCA, Pancatantra exhibition and analysis of children's responses. cite turn0search0 turn0search24

IGNCA, institutional Panchatantra background. cite turn0search13

ElevenLabs, Instant Voice Cloning documentation. cite turn0search3

ElevenLabs, Professional Voice Clone ownership and verification documentation. cite turn0search1 turn0search7

ElevenLabs, Prohibited Use Policy. cite turn0search9

Appendix: CSV starter schema

story_id, title, tantra, source_reference, original_text, translation_en, translation_hi, summary, characters, themes, moral, learning_objective, word_count, sentence_count, vocabulary_complexity, plot_complexity, moral_complexity, comprehension_demand, emotional_sensitivity, cultural_context, preliminary_age_min, preliminary_age_max, validator_rating, sensitivity_flags, review_status

Core research principle

The LLM should not decide what is authentic. The source corpus is the source of truth. RAG retrieves the source, the adaptation engine changes language and presentation for the target child, and validation checks that the moral, events and cultural context remain faithful.